

Expected Uses of Probability Lemma 3.2

Lemma: For any positive integers n and k ,

$$\binom{n}{k} \leq \frac{1}{e} \left(\frac{en}{k} \right)^k.$$

Proof:

Proof. We begin with

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}.$$

Since each factor in the numerator is at most n ,

$$n(n-1)(n-2)\cdots(n-k+1) \leq n^k.$$

Thus,

$$\binom{n}{k} \leq \frac{n^k}{k!}.$$

It remains to bound $k!$ from below. Taking logarithms gives

$$\ln(k!) = \ln 1 + \ln 2 + \cdots + \ln k.$$

Since $\ln x$ is increasing on $[1, \infty)$, we have

$$\ln 1 + \ln 2 + \cdots + \ln k \geq \int_1^k \ln x \, dx.$$

Therefore,

$$\ln(k!) \geq \int_1^k \ln x \, dx = [x \ln x - x]_1^k = k \ln k - k + 1.$$

Hence

$$k! \geq e^{k \ln k - k + 1} = e \left(\frac{k}{e} \right)^k.$$

So

$$\frac{1}{k!} \leq \frac{1}{e} \left(\frac{e}{k} \right)^k.$$

Combining this with $\binom{n}{k} \leq \frac{n^k}{k!}$, we get

$$\binom{n}{k} \leq n^k \cdot \frac{1}{e} \left(\frac{e}{k}\right)^k = \frac{1}{e} \left(\frac{en}{k}\right)^k.$$

Therefore,

$$\boxed{\binom{n}{k} \leq \frac{1}{e} \left(\frac{en}{k}\right)^k}.$$

□